



Does Easy Accessibility to Urban Parks Always Raise Home Values?

Lu Liu^{a,b}, Lina Meng^{c,*}, Ruige Zhang^d

^a School of Management, Guangdong University of Technology, Guangzhou 510006, Fujian, China

^b State Key Laboratory of Urban Land Resources Monitoring and Simulation, Ministry of Natural Resources, Guangdong, 518000, China

^c School of Economics & The Wang Yanan Institute for Studies in Economics, Xiamen University, Xiamen 361005, Fujian, China

^d The Wang Yanan Institute for Studies in Economics, Xiamen University, Xiamen 361005, Fujian, China

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ABSTRACT

This paper employs a boundary discontinuity design to separate the cost of easy access (EA) to urban parks from the combined effects of proximity.¹ On average, houses with EA to urban parks experience a significant price reduction of 2.0%. This reduction equals approximately \$7,986 per house in 2018 dollars, all else being equal. The average negative effect originates from EA to city parks. Mechanism analysis using novel cellphone data reveals that frequent visits to areas with EA to urban parks can lead to congestion disamenities that have a negative effect on the prices of EA houses.

1. Introduction

Urban parks are an important local public good that enhances a city's attractiveness and improves the quality of urban life (Albouy et al., 2020; Angelovski et al., 2018; Jones and Goodkind, 2019; Liu et al., 2020). Moreover, there is evidence that neighborhoods with parks present higher property values (Franco and Macdonald, 2018; Tyrväinen and Miettinen, 2000). While a substantial body of literature exists on the overall effects of proximity to urban parks (Fan et al., 2016; Panduro et al., 2018; Walls et al., 2015), accurately estimating these effects necessitates the separation of negative externalities (such as crime, congestion, air pollution, and noise induced by increased traffic toward urban parks) from positive externalities (such as recreational and aesthetic benefits), which is notoriously difficult.

This paper provides novel empirical evidence on the cost of proximity to urban parks in the following ways. Using a boundary discontinuity design, we separate the net effect of proximity to urban parks on house prices from other factors that are spatially correlated with proximity to urban parks, thereby revealing a significant negative impact of easy access (EA) to urban parks on house prices.² On average, houses with EA to urban parks experience a value reduction of 2.0%, equivalent to approximately \$7,986 per house in 2018 prices. The negative effect of

EA to urban parks is robust after we exclude the impacts of some different public services around the town or county border, noise from major roads, school quality, and taste-based sorting. Furthermore, our findings demonstrate that the negative impacts of EA to urban parks persist over time. By leveraging anonymous cellphone data, we also provide compelling evidence of a significant influx of visitors to areas with EA to urban parks, resulting in disamenities and, consequently, detrimental effects on EA houses.

Our paper contributes to the literature in two ways. First, we provide a novel estimation procedure that separates the net effect on house prices of EA to urban parks, which are spatially correlated with other factors. Previous research has predominantly relied on the distance to urban parks to measure the overall impact of proximity, which combines both benefits and costs (Franco and Macdonald, 2018; Rouwendal et al., 2017). Using a 10-meter digital elevation model (DEM) to measure views, Walls et al. (2015) separate the passive aesthetic benefits of urban greenness from more active use-oriented values. Leveraging spatially detailed data on housing properties and urban parks in Shenzhen, we separate the disamenities of urban parks by a boundary discontinuity design. Our findings reveal a significantly negative effect of EA to urban parks, resulting in a 2.0% decrease in housing value. This effect is different from previous results, which typically indicated combined

* Corresponding author at: School of Economics & The Wang Yanan Institute for Studies in Economics, Xiamen University, Xiamen 361005, Fujian, China.

E-mail addresses: luliuh.wh@gmail.com (L. Liu), lnmeng@xmu.edu.cn (L. Meng), zhangruige@stu.xmu.edu.cn (R. Zhang).

¹ We define a house with EA to urban parks as a given house located on the park side of a major road, within a walking distance of d meters, which will be further discussed in Section 2.1.

² We define a house with EA to urban parks as a given house located on the park side of a major road, within a walking distance of d meters, which will be further discussed in Section 2.1.

effects of proximity to urban parks with positive coefficients ranging from 0.2% to 2% (Franco and Macdonald, 2018; Walls et al., 2015). To our knowledge, this paper represents the first attempt to quantitatively measure the net welfare effect attributable to EA to urban parks. These novel findings should be carefully considered by local governments when designing urban parks.

Second, this paper also contributes to the field of hedonic estimation by incorporating local amenities into housing prices, building upon the works of Rosen (1974). Unobserved neighborhood characteristics, which significantly affect housing prices, are often spatially correlated with the amenities of interest. This correlation introduces potential bias in estimations, challenging hedonic identification. To address the endogeneity caused by time-invariant unobserved characteristics, Black (1999) restricted the estimation sample to a narrow range of boundary discontinuities within school zones. Numerous studies have employed this boundary discontinuity approach to estimate the capitalization into housing price of various amenities, such as school value (Bibler and Billings, 2020; Chan et al., 2020; Gibbons et al., 2013), environmental risk (Christensen et al., 2022; Currie et al., 2015; Muehlenbachs et al., 2015), damage caused by climate change (Bakkensen and Ma, 2020; Bernstein et al., 2019; Bosker et al., 2018), and other disamenities (Gibbons, 2015). Leveraging a transaction-level dataset with detailed geographic information, we employ a boundary discontinuity strategy to quantify the disamenities of EA to urban parks.

This paper proceeds as follows. Section 2 outlines the econometric strategy to separate the disamenities of urban parks from the combined effects. Section 3 describes the data with various spatial information. Section 4 presents regression results accompanied by various robustness checks, followed by further heterogeneity analyses. Potential mechanisms underlying the observed effects are discussed in Section 5. Finally, Section 6 concludes the paper.

2. Econometric Strategy

2.1. Measuring the costs of proximity to urban parks

There is a broad body of literature examining the effects of proximity to urban parks on housing prices. On the one hand, high importance is placed on the aesthetic benefits of proximity to urban parks. Housing values are thus higher in areas with a nearby park (Daams et al., 2016; Tyrväinen and Miettinen, 2000; Walls et al., 2015). Urban parks provide a great aesthetic view and contribute to health effects (Bertram and Rehdanz, 2015; Jones and Goodkind, 2019; Liu et al., 2020). On the other hand, if the distance to urban parks is too small, negative effects, including noise, traffic congestion, fear of crime, or concerns about privacy loss, may arise due to the high volume of visitors (Albouy et al., 2020; Bertram and Rehdanz, 2015; Fan et al., 2016; Kuwayama et al., 2018). Following the literature, we first measure the combined effects of proximity to urban parks using the following specifications:

$$\log P_{ijt} = \beta_0 + \beta_1 D_{ij}^d + \rho X_{ij} + \eta_t \times v_c + q_{qt} + \varepsilon_{ijt} \quad (1)$$

where P_{ijt} is the sale price of house i , located in residential complex j in year t . Different from the Euclidean distance used in the literature (Cohen et al., 2019; Fan et al., 2016; Franco and Macdonald, 2018), we measure the proximity to urban parks by the walking distance (d meters) to the nearest park.³ That is, D_{ij}^d is a dummy, which equals 1 if the walking distance of house i to the nearest park k is less than d meters. The vector $X_{ij,t}$ consists of observable housing and neighborhood characteristics, including the number of bedrooms, logarithmic value of

built-in area, logarithmic value of house age, logarithmic value of lot size, building orientation, a dummy of house type, primary school level, logarithmic value of the distance to the central business district (CBD), logarithmic value of the distance to the nearest metro station, logarithmic value of the distance to the coastline, and number of parks within the control band. The town-by-year fixed effects are represented by $\eta_t \times v_c$, capturing the effect of time-varying local conditions on price P_{ijt} .⁴ Last, q_{qt} is a temporal fixed effect indicating the quarter of the transaction, and ε_{ijt} is the error term. However, as previously discussed, the impacts of housing proximity to urban parks include the benefits of aesthetic views, recreational opportunities, health, and well-being, and the costs of noise, congestion, crime, and privacy concerns. The simplistic specification in Eq. (1) combines the impacts of the benefits and costs of proximity to urban parks by β_1 .

In this paper, we employ a boundary discontinuity approach to separate the benefits and costs of proximity of urban parks from the combined effects. First, we define the dummy *Access*, which identifies houses on the park side of a geographic obstacle, i.e., a major road.⁵ Both D^d and *Access* are time-constant dummy variables. Second, *Open* is a dummy that takes a value of one if the transaction is made after the park is built. Finally, we define EA to urban parks as being located on the park side of a major road, within a walking distance of d meters. That is, EA is the interaction of ($D_{ij}^d \times Access_{ij} \times Open_{ijt}$). Typically, areas with EA to urban parks bear more visitors and may suffer from noise, congestion, crime, etc. (Albouy et al., 2020; Kuroda and Sugawara, 2023; Liu et al., 2020). We then use the difference in differences in the average housing value of having access to the park area versus not having access and being within the walking distance band compared to the differences in the average house value of having access versus not having access to the park area and being in the control band that is further from the park in the same years after versus before the park opens, to measure the costs of houses with EA to urban parks.

Fig. 1 describes the measurement of costs and other benefits of a given house near an urban park. The first distance band covers the areas from the entrance gate of a given urban park to a walking distance of d meters (i.e., 800 m), depicted by the orange and yellow areas in Fig. 1 (areas A and B). We assume that houses located within this distance band and on the park side of a major road (areas A in Fig. 1) may suffer additional costs of noise, congestion, fear of privacy loss, etc. (Kuroda and Sugawara, 2023).⁶ According to the urban planning guidelines in Shenzhen,⁷ a major road is a two-way road with a median strip, serving as a geographical barrier that limits access from the park side to the other side (area B in Fig. 1). Consequently, individuals living in houses located within area B enjoy the benefits of proximity to the park while experiencing fewer associated costs.⁸ The sale prices of these houses reflect the capitalization of park benefits, albeit with a lesser discount on costs. Moreover, houses located within a walking distance band of [$d+200, 2d+200$] m on the park side of a major road capture the effects

⁴ A town is the lowest administrative level in China, and the median size of a town in Shenzhen is approximately 19.18 square km.

⁵ See Fig. A1 in the Appendix A as a conceptual graph of a major road.

⁶ According to statistics, the cases accepted for processing represent 2.97 per thousand residents in 2008, while the number is slightly reduced to 2.74 in 2019. Moreover, the crime of property infringement, including robbery and mugging, decreased by 92.44% from 2008 to 2019. Similarly, violent crime decreased by 23.98%, resulting in 0.25 violent crimes per thousand residents in 2019. Given the relatively small number of crimes in Shenzhen, we conclude that the costs of crime due to proximity to urban parks are minor in our study. Data source: https://m.thepaper.cn/newsDetail_forward_3801966.

⁷ It is "Shenzhen Urban Planning Standards and Guidelines (2004)" approved by the Shenzhen Municipal People's Government in 2004, which was updated in 2014 and 2021.

⁸ We measure the benefits of proximity to urban parks by walking distance. The benefits of houses in areas A and B are thus comparable.

³ The walking distance refers to the distance from the house to the nearest park entrance. We measure the walking distance by ArcGIS using the road networks from Open Street Map: <https://www.openstreetmap.org/#map=10/22.5608/114.1477>.

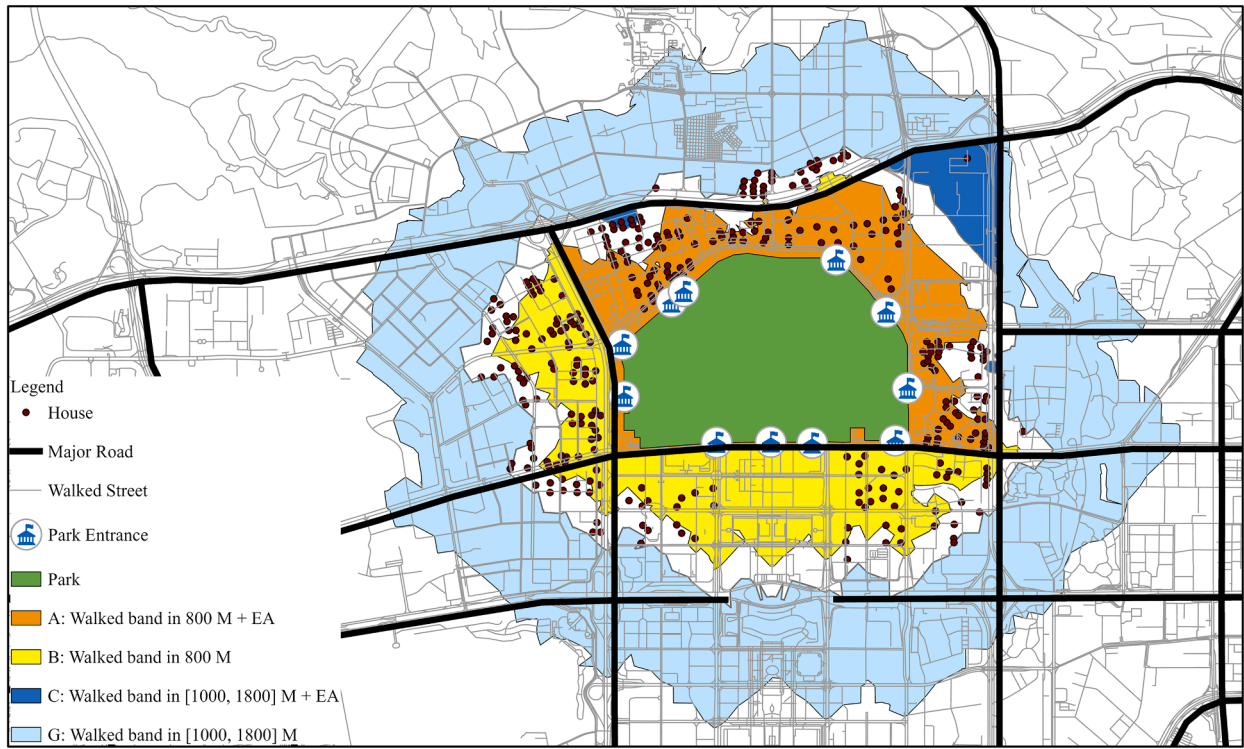


Fig. 1. Sample of within-band variation in EA to a park.

Note: This figure displays the locations of houses based on their proximity to urban parks. The four categories are as follows: (A) houses within an 800 m walking distance and located on the park side of a major road, (B) houses within an 800 m walking distance but located on the opposite side of a major road, (C) house within a 1000 - 1800 m walking distance and located on the park side of a major road, and (G) houses within a 1000 - 1800 m walking distance but located on the opposite side of a major road. The bold lines are major roads. The gray lines are the walked street network sourced from Open Street Map.

of being adjacent to the park but in a relatively remote area (areas C in dark blue in Fig. 1). To provide a counterfactual sample, we use nearby house transactions as controls. That is, houses located within walking distance of $[d+200, 2d+200]$ m and on the opposite of the park side of a major road (areas G in light blue in Fig. 1) serve as controls.⁹

To conduct an empirical estimation using the preceding intuition, we implement a boundary discontinuity design:

Referring to Fig. 1 and taking houses in area G before the park opens as the control group, the price changes in houses in areas A, B, C and G, after versus before the park opens, are as follows:

$$\begin{aligned}\Delta P_A &= \beta_2 + \beta_4 + \beta_5 + \beta_6 \\ \Delta P_B &= \beta_2 + \beta_4 \\ \Delta P_C &= \beta_2 + \beta_5 \\ \Delta P_G &= \beta_2\end{aligned}$$

$$\begin{aligned}\log P_{ij,t} &= \beta_0 + \beta_1 D_{ij}^d + \beta_2 Open_{ij,t} + \beta_3 Access_{ij} + \beta_4 D_{ij}^d \times Open_{ij,t} + \beta_5 Access_{ij} \times Open_{ij,t} \\ &+ \beta_6 (D_{ij}^d \times Access_{ij} \times Open_{ij,t}) + \rho X_{ij,t} + k_{ij, Park} + \eta_t \times v_c + q_{qt} + \varepsilon_{ijt}\end{aligned}\quad (2)$$

where $P_{ij,t}$ is the sale price of house i , located in residential complex j in year t . The dummy variable $Access_{ij}$ takes a value of one if house i located on the park side of a major road.¹⁰ $Open_{ij,t}$ takes a value of one if the transaction happens after the park k built. The park fixed effects $k_{ij, Park}$ ensure that the comparisons are among houses nearest to the same park. All other variables are the same as defined in equation (1).

⁹ To eliminate spillover effects, we exclude houses within walking distances of $(d, d+200)$ m in the estimations. See our discussions in Section 4.1. Moreover, the distance between the park and the major road that separates the first distance bank into areas A and B varies for each park. Therefore, our estimation strategy captures the average impact of EA to urban parks.

¹⁰ To avoid any potential confounding effects arising from the construction of a new major road on the park side, we use samples before and after the park opens, but excluding parks that opened before the constructions of the major roads.

this implies the following:

$$(\Delta P_A - \Delta P_B) - (\Delta P_C - \Delta P_G) = [(\beta_2 + \beta_4 + \beta_5 + \beta_6) - (\beta_2 + \beta_4)] - [(\beta_2 + \beta_5) - \beta_2] = \beta_6$$

thus, β_6 is the triple-difference measure of the impacts of EA to urban parks; that is, the difference in the differences in the average housing value of having access to the park area versus not having access and being in the 0–800 m walking distance band compared to the differences in the average house value of having access versus not having access to the park areas and being in the 1000–1800 m walking distance band in the same years after versus before the park opens, all other being equal. In this way, we separate the benefits and costs of proximity to urban parks. β_4 is the difference-in-differences in the prices of houses located in the walking distance band of 0–800 m and on the opposite side of a major road in the same years after versus before the park opens,

compared to those located in the control band of [1000, 1800] m to the same park and on the opposite side of a major road in the same years.

It is possible that price discontinuities exist across major roads not only due to cost disparities but also due to variations in other characteristics. A typical example is where major roads coincidentally partition towns that offer different local public goods (Gibbons et al., 2013). To examine the robustness of our estimation, we further exclude houses within a 100 m buffer area of each town boundary in the robustness tests.¹¹ Furthermore, we exclude houses within a 100 m buffer area of a major road to remove the impacts of noise from the major road in the robustness tests (von Graevenitz, 2018).¹²

2.2. The threat of sorting

Despite the spatially detailed boundary discontinuity strategy employed above, there remains a potential concern of sorting bias across different border zones. The effectiveness of boundary discontinuity design relies on the assumption that individuals are unable to strategically manipulate their location relative to the boundary (Gibbons et al., 2017). The issues of income or taste-based sorting across geographical boundaries are pervasive in boundary discontinuity strategies. One common way to address this problem is to control for observable characteristics (Bayer et al., 2007) or use proxy measurements for an individual's preferences, which are often unobservable. We address the sorting of households by the idea that if people sort into areas with different accessibility to urban parks based on observable characteristics, these characteristics should be systematically related to different accessibility to urban parks (Bosker et al., 2018; Greenstone and Gallagher, 2008). Specifically, relying on 2015 mini-census data in Shenzhen, we match household characteristics to housing sales according to addresses from both datasets. Subsequently, we replace the dependent variable in Eq. (2) with either household demographic characteristics or job status.

3. Data and Descriptive Analysis

3.1. Datasets

The analysis of this study is mainly based on four datasets. The first dataset contains housing sales from January 2005 to April 2018. This dataset is sourced from four reputable real estate brokers in Shenzhen, one of the most expensive cities in China. It covers approximately 50% of the transactions in Shenzhen and provides crucial information on each sale, including transaction price, address, and specific details about the house, such as living area, number of bedrooms, number of washrooms, year of construction, building head, and type of building. We link the transaction records to the building census dataset according to the unique building ID to obtain the geographical location (latitude and longitude) of each transaction.

Location is one of the most important factors affecting housing prices. We calculate a series of location-related variables by geographic information system (GIS), including distances to key landmarks such as the central business district (CBD), shopping malls, metro stations, the coastline, hospitals, and major roads. The quality of the primary school

attendance zone in 2018 is used as a proxy for the school quality of a house in our sample.¹³

The second dataset contains information on the entrance gates and spatial boundaries of urban parks constructed from 1982 to 2018, acquired from the Shenzhen open data platform.¹⁴ This dataset also contains information on the attributes of each park, including the name, area, year of construction, and park type. There are three types of parks in Shenzhen: neighborhood parks, city parks, and national parks. These classifications are based on the "Classification Standards for Urban Green Spaces (2017)" in China, which consider factors such as service radius and intended purpose. Neighborhood parks primarily cater to local residents and offer amenities for activities such as sports, walks, games, and leisure. They are typically shared by multiple residential communities, with an average distance of 492 m to nearby residential complexes. In contrast, city parks are designed to fulfill the recreational needs of residents throughout the entire city. They provide spaces for rest, sightseeing, exercise, social interaction, and various cultural activities. The mean distance from residential complexes to the nearest city park is 1316 m. There are 23 national parks in Shenzhen. They are primarily designed for nature conservation and are located far from residential complexes, with an average distance of 11.72 km. We thus exclude national parks from our study. Fig. A2 in Appendix A provides images of neighborhood parks and city parks. We further limit parks to those with house transactions within $2d + 200$ m, resulting in 151 parks in the final sample, 84 of which were built after 2005. Table A2 in Appendix A provides the summary statistics of the park characteristics, grouped by parks' construction years. The number of parks, including both neighborhood and city parks, has increased over time, with city parks having larger average areas than neighborhood parks. Fig. 2 depicts the locations of the houses sold, the spatial boundaries of the urban parks, and the road networks included in the study. Newly constructed parks are generally located away from the city center. Fig. A3 in Appendix A provides additional information by displaying the mean distance of parks to the city center of Shenzhen. Notably, the mean distance of neighborhood parks to the city center remains relatively consistent over time, while the mean distance of city parks increases substantially. That is, the newly built city parks are generally located in peripheral areas, while the locations of the newly built neighborhood parks are distributed more evenly in both core and peripheral areas of the city.

The third dataset is the national 1% population sample survey from 2015 (hereafter referred to as the 2015 mini-census data), which provides information on household demographic characteristics, living conditions, work status, etc. We geocode the 2015 mini-census data through the address of each household and then calculate the walking distance as well as the accessibility to urban parks by the same methods described in Section 2.1. Overall, the 2015 mini census data consists of 4309 households in Shenzhen, of which 3664 households are successfully geocoded and matched to other data by location.

In the mechanism analysis, we also use anonymous cellphone data to measure individuals' park visits. The cellphone dataset adopts a $250\text{ m} \times 250\text{ m}$ hexagon as the unit of analysis. In October 2019, there were 31,487 hexagons in Shenzhen, capturing 56 million residents' location and mobility.¹⁵ The operator provides some users' information and mobility, aggregated to the hexagon level. Fig. A4 in Appendix A visually presents how cellphone data measures residents' mobility. By overlaying the hexagons with the urban parks and major road layers in

¹¹ The mean distance from a house to the nearest towns boundaries is approximately 800 meters. The elimination of houses within a 100-meter buffer of community/town boundaries results in dropping 10.29% of the samples (23,258 property transactions).

¹² Supposing a major road with 55 m wide is located 500–555 m from a park, the elimination of a 100 m buffer area of a major road would divide the inner walking distance bands into 0–400 m and 655–800 m.

¹³ The school attendance zone may change over time. Unfortunately, we have the zone map for only 2018. We exclude samples for which the school attendance zone changed in the robustness analysis.

¹⁴ The website of the platform: <https://opendata.sz.gov.cn/>. There are 791 parks in the dataset.

¹⁵ There were 134.38 million residents in Shenzhen in 2019. This anonymized cellphone dataset covers approximately 40% of the population.

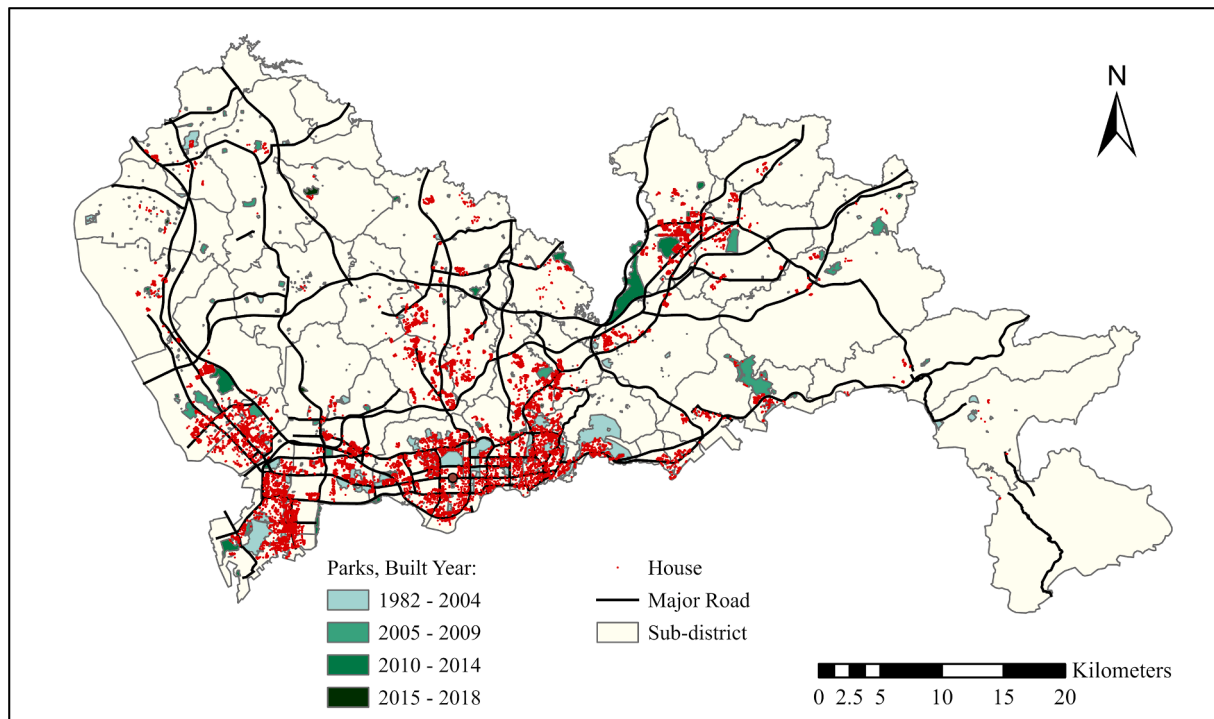


Fig. 2. Survey areas: urban parks and locations of transactions.

Note: The figure shows the spatially detailed dataset used in this study. The red dots represent the locations of houses. The dark green areas are newly constructed parks, while light green areas are former urban parks in Shenzhen. Bold lines depict the major roads in Shenzhen. The major roads have a minimum speed limit of 40 km/h and a width of more than 55 m.

ArcGIS, we narrow down the sample to those whose destination hexagon is within the $2d+200$ m walking distance of an urban park.

3.2. Descriptive Analysis

Table 1 provides the descriptive statistics of the variables used in the estimation. Houses in the sample are restricted to those within an 1800 m walking distance of the nearest park. Overall, there are 87,943 transactions in the sample, and the average price of houses in Shenzhen is \$388.92 thousand (2018 prices). More than 50% of the houses are located within an area of 800 m walking distance to the nearest park, and approximately 64% of them are located on the park side of the major road. Approximately 95% of the transactions occurred after the park was built. We also summarize various observable housing characteristics in Table 1, including house age, living area, number of bedrooms, and size of the residential complex. On average, houses were sold at approximately 10 years of age, with living areas typically less than 80 square meters and an average of 2.25 bedrooms, and the average size of a residential complex was 2.76 hectares (ha). Table A1 in Appendix A also provides summary statistics of the entire dataset.

Panel B of Table 1 provides the summary statistics of the 2015 mini-census data. On average, within a household, approximately 2% were children younger than 7 years old, while 4% were aged 60 years or older. Approximately 50% of household members held a high school diploma and higher, and 12% were unemployed. The degree of social mixing, measured by the percentage of household members with a Shenzhen hukou, varied among families. Overall, within households, 16% were residents with Shenzhen hukou.

Table 2 further provides summary statistics of the cellphone data. On average, 942.50 users visited hexagons close to parks per day during October 2019 (Column (4) in Table 2). As expected, more visitors intentionally visited areas with EA to urban parks (approximately 1652.60 visitors per day) compared to areas on the opposite side of a major road (905.87 visitors per day on average). This dataset also

provides information on the number of people who live or work within a specific hexagon, according to the duration of their stay.

4. Results

4.1. Searching around the distance bands

We first use the regression model of Eq. (1) to estimate the treatment effects of proximity to urban parks on housing prices within various distance bands. The results in Fig. 3 show that, conditional on a set of observable attributes, housing prices increase by approximately 2%. The effects of proximity to urban parks disappear when the walking distance from a house to the nearest park is more than 800 m. Overall, the patterns in Fig. 3 suggest that, the estimated effects become statistically insignificant beyond 800 m. These findings in Fig. 3 provide supportive evidence that an 800 m walking distance is a valid cutoff distance to define the treatment group. To eliminate the potential spillover effects, we employ a “doughnut” boundary discontinuity specification, as suggested by Michaels et al. (2021). Specifically, houses within a walking distance of 800–1000 m are excluded from the estimations. Moreover, to ensure the comparability of house attributes, we set the control group as houses within 1000–1800 m walking distance to the nearest park. Furthermore, an event study in Appendix B1 provides suggestive evidence that the parallel pretrends of housing prices between houses with EA and those without EA hold.

4.2. Baseline estimations: net effect of EA to urban parks on housing prices

Relying on the cutoff treatment buffer of 800 m walking distance shown in Fig. 3; Table 3 presents our baseline estimations of the average effects of EA to urban parks on housing prices. This approach provides a brand-new perspective on the net effects of EA to an urban park rather than the combined effects of proximity to urban parks. Specifically, we

Table 1
Descriptive statistics.

	N	Mean	S.D.	P1	P99
Panel A: Houses characteristics					
Price (thousand \$)	87,943	388.92	345.62	483.35	1,880,070
D	87,943	0.54	0.5	0	1
Access	87,943	0.64	0.48	0	1
Open	87,943	0.95	0.23	0	1
Dist. to the nearest parks (m)	87,943	870.5	467.4	31.94	1762.00
Area of the nearest park (ha)	87,943	11.47	1.65	9.22	15.51
No. of parks	87,943	0.28	0.53	0	2
Age of house	87,943	10.82	5.51	1	26
Living areas (m ²)	87,943	77.49	32.23	24.11	187.80
No. bedroom	87,943	2.25	0.93	1.00	5.00
Area of residential complex (ha)	87,943	2.76	3.36	2.05	160.00
Primary school level	87,943	1.27	1.49	0	4
Dist. to CBD (km)	87,943	10.33	6.79	0.06	32.55
Dist. to mall (km)	87,943	1.42	1.50	0.08	9.48
Dist. to metro station (km)	87,943	1.53	2.49	0.06	12.46
Dist. to coastlines (km)	87,943	5.75	4.38	0.28	15.76
Dist. to hospital (km)	87,943	1.46	1.07	0.10	4.90
Dist. to major road (m)	87,943	185.20	146.90	34.18	733.20
Panel B: Demographic characteristics (2015 mini census)					
Age between 0 - 6	3664	0.02	0.08	0	1
Age above 60	3664	0.04	0.17	0	1
Highly educated	3664	0.49	0.46	0	1
Unemployed	3664	0.122	0.28	0	1
Social mixed	3664	0.16	0.36	0	1

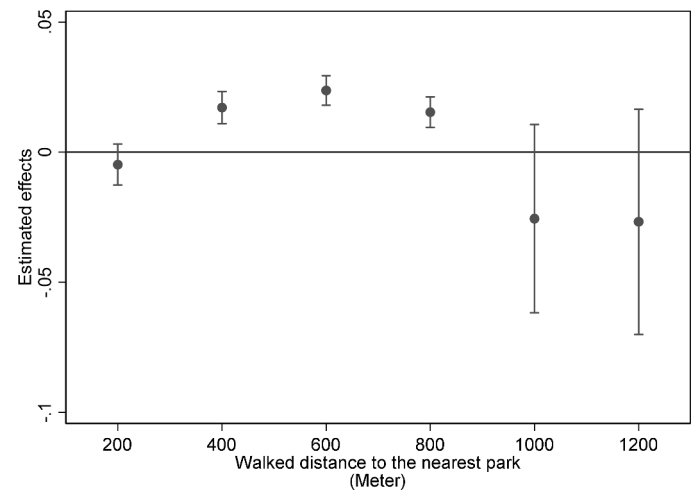
This table includes summary statistics of houses sold from January 2005 to April 2018. Houses in the sample are restricted to those within an 1800 m walking distance of the nearest park. The housing prices data are converted to 2018 real dollars (\$) according to the exchange rate of 6.6174 RMB/\$. Distance to a park is measured by walking distance. The dummy *D* takes a value of one if a house is located within or equal to an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. *Access* is a dummy that takes a value of one if a house is located on the park side of a major road, within a walking distance of 0 – 1800 m to the nearest park. *Open* is a dummy that takes a value of one if the transaction is made after the park is built. *Primary school level* is a categorical variable; the primary school with the highest quality is coded with a value of 4, while a school with the lowest quality is coded with a value of 0. *Highly educated* is the percentage of people holding a high school diploma or above in a household. *Social mixed* is the percentage of people having local hukou in a household.

measure the triple-difference between EA houses and houses within the 800 m walking distance but not on the park side in the same years after versus before the park opens, as well as between houses on the park side and within the control distance band of 1000–1800 m and houses in the same control distance band but not on the park side in the same years after versus before park opens.

Table 2
Summary statistics of the anonymized cellphone data.

	N (1)	Mean (2)	S.D. (3)	D = 1 Mean (4)	D=0 Mean (5)	Access =1 Mean (6)	Access =0 Mean (7)
# visitors per hexagon pair	23,278,561	0.42	5.03	0.44	0.39	0.45	0.41
# visitors per destined hexagon	32,790	940.15	1942.14	942.50	932.03	1652.60	905.87
<i>D</i>	32,790	0.47	0.50	1	0	0.75	0.43
<i>Access</i>	32,790	0.13	0.34	0.21	0.06	1	0
<i>Home</i>	32,790	883.40	1114.17	1405.31	883.40	1452.32	798.19
<i>Work</i>	32,790	938.56	1225.35	1415.57	938.56	1404.14	868.83

This table summarizes the daily visits to urban parks measured by anonymized cellphone data. Only the destination hexagons within the walking distance band of [0, 800] m and the control distance band of [1000, 1800] m to urban parks are included. The size of each hexagon is 250 m × 250 m. # Visitor per hexagon pair refers to the daily mean number of people moving from hexagon *s* (origin) to hexagon *h* (destination). # visitors per destination hexagon *h* is the total daily visitors to destination hexagon *h*. *D* is a dummy, which takes a value of one if destination hexagon *h* is located within a walking distance band of [0, 800] m to urban parks and zero otherwise. *Access* is a time-constant dummy that takes a value of one if hexagon *h* is located on the park side of the major road at the end of the study period and zero otherwise. *Home* measures the number of people who live in a given hexagon, while *work* measures the number of people who work in a given hexagon.

**Fig. 3.** The effects of proximity to the nearest parks on housing prices.

Note: This figure depicts the effects of proximity to the nearest parks on housing prices. The sample used in the analysis is all sales after park opening and within a walking distance of [0, 2400] m to the nearest park. Each dot is based on a separate regression of housing prices on a walking distance band *d* meters. The vertical lines represent 90% confidence intervals. The observable housing characteristics are defined in Table 1, and a full set of town-year fixed effects is included in each regression.

In Column (1) of Table 3, we simply regress the proximity indicator *D* (parks within 800 m) on housing prices (in logarithmic form). Similar to the results in Fig. 3, the result suggests a significant positive effect of proximity to urban park areas on house prices, which combines the effects after and before the park opened. Column (2) adds the indicator *Open* and the interaction term to the regression. The results show that park opening (*Open*) significantly decreases housing prices by 1.5% in the control band. Moreover, the interaction term of *D* and *Open* remains significantly negative, indicating that parks are not always good. Parks even have negative effects when their disamenities, such as increased noise and traffic and fear of privacy loss outweigh their amenities (Kuroda and Sugawara, 2023).

Column (3) further includes the triple-difference estimator to separate the effects of EA to urban parks within the walking distance on housing prices, as illustrated by Eq. (2). The results suggest that houses located in a walking distance band of 0–800 m and on the opposite side of a major road in a year before the park opens are 4.1% higher compared to the prices of houses in the walking distance band of 1000–1800 m and on the opposite side of a major road on the same years (houses in area G in Fig. 1), on average, all else being equal. Additionally, the difference-in-differences in the prices of houses located in the walking distance band of 1000–1800 and on the park side of a major

Table 3
Baseline results.

	(1)	(2)	(3)
D	0.010*** (0.003)	0.034** (0.008)	0.041*** (0.008)
Open		-0.015** (0.006)	-0.020** (0.010)
D × Open		-0.025** (0.008)	-0.023** (0.009)
Access			-0.035*** (0.011)
Access × Open			0.059*** (0.011)
D × Access × Open			-0.020*** (0.006)
Control variables	Yes	Yes	Yes
Park FE	Yes	Yes	Yes
Town × Year FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
N	87898	87898	87891
adj. R ²	0.9	0.899	0.908

This table reports the baseline results comparing houses with and without EA. The dependent variable is the logarithmic form of the transaction price. The dummy *D* takes a value of one if a house is located within an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. *Access* is a dummy that takes a value of one if a house is located on the park side of a major road, within a walking distance of 0 – 1800 m to the nearest park. *Open* is a dummy that takes a value of one if the transaction is made after the park is built. Each regression controls for the number of bedrooms, logarithmic value of the built-in area, logarithmic value of house age, logarithmic value of lot size, building orientation, a dummy of house type, primary school level, logarithmic value of distance to the central business district (CBD), logarithmic value of distance to the metro station, logarithmic value of distance to the coastline, and number of parks within 2d distance band, park fixed effects, town × year fixed effects, and quarter fixed effects. Robust standard errors are in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

road from after versus before the opening of the park and prices of houses in the walking distance band of 1000–1800 m and on the opposite side of a major road in the same years is 5.9%. However, the EA to urban parks causes a significant decrease in housing prices amounting to 2.0%, on average, all else being equal.¹⁶

Prior to conducting several robustness checks on our baseline estimation, we employ back-of-the-envelope analysis to estimate the economic loss of houses with EA to urban parks. The calculation process is as follows. For houses on the park side of a major road, we use the estimates in Column (3) of Table 3 to predict the price discount from the mean total value of the houses sold. Next, we amortize the total value loss into a 30-year period of residence to obtain the annual loss associated with EA to urban parks (Chu et al., 2020). We use the Chinese compound interest rate for five-year fixed deposits (2.75% per year) and five-year treasury bills (3.49% per year) in 2018 as the lower and upper bounds of the discount factors, respectively. With these numbers, we estimate that the total average value loss of houses with EA to urban parks is \$7.986 thousand (2018 prices) per house, and the annual loss is approximately \$247 - \$256.

In summary, our baseline estimation shows that houses with EA to urban parks experience an average price discount of 2.0%. This result differs from the results in previous literature, which addressed the combined effects of proximity to urban parks, as estimated by Franco and Macdonald (2018) and Walls et al. (2015). The average economic loss of a house with EA to urban parks is approximately \$247 - \$256 per year.

¹⁶ We do not have data on the land use types of park areas before they were officially designated as parks. If a park area was a disamenity beforehand, then the net benefits would likely be higher than if it was an amenity (e.g. hazardous waste site versus an open space that was not well maintained).

Table 4
Robustness checks.

	Excluded sub-district border sample (1)	Excluded road border sample (2)	Excluded school border changed sample (3)
D	0.042*** (0.008)	0.030*** (0.012)	-0.003 (0.020)
Access	-0.031*** (0.011)	-0.059*** (0.014)	-0.066** (0.023)
Open	-0.017* (0.010)	-0.047*** (0.011)	-0.054** (0.019)
D × Open	-0.021** (0.010)	0.013 (0.013)	0.004 (0.022)
Access × Open	0.054*** (0.011)	0.078*** (0.014)	0.089*** (0.022)
D × Access × Open	-0.025*** (0.006)	-0.045*** (0.009)	-0.053*** (0.012)
Control variables	Yes	Yes	Yes
Park FE	Yes	Yes	Yes
Town × Year FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Excluded town border sample	Yes	Yes	Yes
Excluded road border sample	No	Yes	Yes
Excluded school border changed sample	No	No	Yes
N	82868	58423	53856
adj. R ²	0.900	0.906	0.906

This table reports the results of various robustness checks. The dependent variable is the logarithmic form of the transaction price. The dummy *D* takes a value of one if a house is located within an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. *Access* is a dummy that takes a value of one if a house is located on the park side of a major road, within a walking distance of 0 – 1800 m of the nearest park. *Open* is a dummy that takes a value of one if the transaction is made after the park is built. Column (1) excludes the samples within 100 m of a town border. Column (2) further excludes the samples within 100 m of a major road, and Column (3) excludes the samples in areas where the school attendance zone changed. Each regression controls park fixed effects, town × year fixed effects and quarter fixed effects. Robust standard errors are in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3. Robustness checks

In this section, we conduct several robustness checks to strengthen our analysis. Notably, major roads often overlap with town or county boundaries, leading to differences in public services for residents of houses with EA. To address this concern, we exclude houses located within 100 m of a town boundary. Column (1) of Table 4 reports the results estimated by Eq. (2). We find robust discount effects of EA on house prices, after removing the samples close to town boundaries.

The measurement of EA may involve the disamenities of noise from major roads. As such, it is possible that EA aggravates the negative effect of noise on housing prices. To account for this, we further exclude houses within 100 m of a major road.¹⁷ The results in Column (2) show that the negative effect of EA on housing prices remains significant.

In China, school quality significantly affects individuals' willingness to pay for a house (Chan et al., 2020). The primary school attendance zone may change over time, which ultimately biases our estimations. To address this concern, we exclude samples that experienced a change in school attendance zone during the studied period, and the results are presented in Column (3) of Table 4. We find that the negative impact of EA on housing prices holds.

¹⁷ von Graevenitz (2018) showed that the road noise decreases sharply within a 100 m distance, and there is no significant variation in noise across distances greater than 100 m.

Table 5

Taste-based sorting.

	% with children (1)	% with the elders (2)	% with highly educated (3)	% with jobless (4)	% with local hukou (5)
D	0.048* (0.028)	0.043 (0.088)	-0.038 (0.026)	0.119 (0.130)	0.008 (0.167)
Open	0.030 (0.178)	-0.001 (0.009)	0.531* (0.279)	0.034* (0.019)	-0.181 (0.182)
Access × Open	-0.063 (0.181)	0.095 (0.064)	-0.764** (0.322)	-0.013 (0.132)	0.267 (0.196)
D × Access × Open	0.020 (0.035)	-0.074 (0.097)	0.260 (0.186)	-0.064 (0.150)	0.025 (0.150)
Control variables	Yes	Yes	Yes	Yes	Yes
Park FE	Yes	Yes	Yes	Yes	Yes
Town FE	Yes	Yes	Yes	Yes	Yes
N	2274	2274	2274	2274	2274
adj. R ²	0.044	0.263	0.203	0.401	0.157

This table reports the evidence of taste-based sorting, estimated by Eq. (2). The dependent variables are demographic structure and social status at the household level. The dummy *D* takes a value of one if a house is located within an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. *Access* is a dummy that takes a value of one if a house is located on the park side of a major road, within a walking distance of 0 – 1800 m of the nearest park. *Open* is a dummy that takes a value of one if the transaction is made after the park is built. Each regression controls the lot-specific control variables. We also include town fixed effects to control for unobservables. Robust standard errors are in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6

Systematic differences in willingness to pay for EA effects in different parks.

	Neighborhood park (1)	City park (2)
D	0.125*** (0.017)	0.017* (0.010)
Access	-0.031 (0.023)	0.042*** (0.014)
Open	0.075*** (0.024)	-0.002 (0.012)
D × Open	-0.152*** (0.018)	0.017 (0.012)
Access × Open	-0.006 (0.023)	0.024 (0.015)
D × Access × Open	0.045*** (0.009)	-0.051*** (0.008)
Controls variables	Yes	Yes
Park FE	Yes	Yes
Town × year FE	Yes	Yes
Quarter FE	Yes	Yes
N	38504	49376
adj. R ²	0.898	0.904

This table reports the heterogeneity in individuals' willingness to pay for the EA effects of different types of urban parks. The dependent variable is the logarithmic form of the transaction price of each property. The dummy *D* takes a value of one if a house is located within an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. *Access* is a dummy that takes a value of one if a house is located on the park side of a major road, within a walking distance of 0 – 1800 m of the nearest park. *Open* is a dummy that takes a value of one if the transaction is made after the park is built. Each regression controls the hedonic relative variables, park fixed effect, town × year fixed effects, and quarter fixed effects. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Finally, we employ 2015 mini-census data to probe whether there is taste-based sorting over the EA to urban parks. Inspired by Greenstone and Gallagher (2008), we explore evidence of sorting by replacing the dependent variables in Eq. (2) with either demographic structure or social status at the household level. Two proxies, the percentage of children less than 7 years old and the percentage of adults over 60 years old in the household, are based on the idea that families with children or elderly individuals have a strong preference for recreation activities in urban parks. Another proxy, i.e., the percentage of highly educated people in the household, is based on the idea that families with highly educated members might be sensitive to noise caused by increased

traffic on the park side of a major road. The social mix is based on the idea that families with members having Shenzhen hukou have more information regarding the benefits and costs of the local amenities, which makes them systematically sort into the side of a major road with more amenities.

The results in Table 5 show that, on average, households with children younger than 7 years old are more likely to choose houses located in a walking distance band of 0–800 m to an urban park, compared to houses located in a control band of 1000–1800 m, all else being equal, with marginal significance at a 10% level. However, the triple-difference estimator of houses with EA to urban parks is positive but insignificant. The results remain insignificant when we replace the dependent variable with other demographic structures and social statuses, including the percentage of 60-year-olds, highly educated people, jobless family members, and people with local hukou. In summary, we find no evidence of taste-based sorting in Table 5.

Our results thus provide robust evidence of the negative effects of houses with EA. Using the boundary discontinuity approach, we find that houses in the treatment buffer with EA to parks experience an average price discount of 2.0%. The results are robust when we exclude samples that may cause confounders in the robustness. We also find suggestive evidence that residents do not systematically sort across borders.

4.4. Systematic differences in the negative effects of EA on housing prices

Individuals' willingness to pay for different types of urban parks may influence the negative effects of houses with EA to a certain extent (Liu et al., 2020). In this section, we begin by examining the heterogeneity in the negative effects of houses with EA to two different types of parks: neighborhood parks and city parks. Generally, city parks are large in area, and developed for multiple purposes. The summary statistics in Table A3 of Appendix A provide initial evidence that the housing characteristics of houses around different types of urban parks are comparable, except for the price and nearest walking distance to the parks. The results show that the negative impacts of EA to parks originate from city parks. Using the boundary discontinuity approach as shown in Eq. (2), the results in Columns (1)–(2) of Table 6 suggest that, on average, houses with EA near city parks suffer a significant discount of 5.1% of their value. Individuals living around neighborhood parks, however, are willing to pay for the EA of these parks. On average, the prices of houses with EA near neighborhood parks are significantly higher, with an increase of 4.5% of their value. It may be inferred that

city parks built for multiple purposes may attract more visitors, causing noise, traffic congestion, etc. (Kuroda and Sugawara, 2023; Liu et al., 2020). All these negative externalities of EA to city parks result in a significant price discount for nearby houses. Neighborhood parks are built for local residents to offer amenities for activities such as sports and leisure, such that the benefits outweigh the costs of EA to these parks. We will provide further evidence to support this argument in the mechanism analysis. Overall, we find that the negative impacts of EA on housing prices originate from city parks. On average, houses with EA to city parks experience an average price discount of 5.1%, while houses with EA to neighborhood parks experience an average price premium of 4.5%.

We next investigate how the negative effect of EA to urban parks varies over time. To do this, we construct a series of dummy variables, $Post_t$ ($t \in [1, 10]$), which equals 1 if a house is sold after t years since the park's opening.¹⁸ The results in Fig. 4 show that the coefficients for $D \times Access \times Post_t$ are statistically negative over time, implying that the negative effects of EA to urban parks persist over time.

The hedonic approach, coupled with boundary discontinuity analysis, provides clear evidence of a negative effect of EA to urban parks on housing prices. We do not find sorting across major roads, which provides supportive evidence that our estimations are unbiased. However, we find some systematic differences in the negative effects of EA on urban parks. Specifically, houses near city parks are more sensitive to the disamenities associated with EA. Additionally, we show that the negative effects of EA to urban parks persist over time.

5. Mechanism analysis

In Section 2, we intuitively summarize the disamenities caused by EA to urban parks, including increased noise and traffic due to the high volume of visitors, and concerns about privacy loss. Due to data limitation, we explore only one of the potential mechanisms behind the negative effect of EA, that of the large volume of visitors. Specifically, we employ anonymous cellphone data on mobility information to explain the potential mechanisms: one of the disamenities of EA to urban parks is caused by a large volume of visitors to the park side of a major road.

First, we regress the logarithmic value of daily visits to a given hexagon h on a series of distance dummies, using the specifications described in Eq. (1). The results in Fig. B2 in Appendix B show that, overall, the number of visits to hexagons within an 800 m walking distance to urban parks is greater than the number of visits to hexagons within a walking distance of 1200–2400 m. We further separate the combined effects of proximity to urban parks on visits by a boundary discontinuity approach:

$$\begin{aligned} \text{Log}(\text{Visits}_{sh}) = & \theta_0 + \theta_1 D_h + \theta_2 \text{Access}_h + \theta_3 (D_h \times \text{Access}_h) + \mu X_h + k_{h,park} \\ & + \varepsilon_{sh} \end{aligned} \quad (3)$$

where $\text{log}(\text{Visits}_{sh})$ is the logarithmic value of daily visits to a given hexagon h , originating in a given hexagon s . The dummy D_h takes a value of one if hexagon h is located within or equal to an 800 m walking distance of the nearest park, and zero if hexagon h is within a 1000–1800 m walked distance of the nearest park, and Access is a time-constant dummy that takes a value of one if hexagon h is located on the park side of a major road, and zero otherwise. The interaction term of $(D_h \times \text{Access}_h)$ thus captures the difference in daily visits to hexagon h

located in a walked band of 0–800 m and on the park side of a major road, compared to hexagons in the control distance band of 1000–1800 m and on the opposite side of a major road.¹⁹ X_h is a vector of controls, including the logarithmic number of users living in hexagon h , the logarithmic number of users working in hexagon h , and the logarithmic distance from hexagon h to the nearest major road. In addition, $k_{h,park}$ is park fixed effects. Note that we have cellphone data spanning only one-month, which limits our comparisons of visits to park areas relative to before the park opening.

The results of the mechanism analysis using Eq. (3) are shown in Table 7. Column (1) shows that hexagons located in the walking distance band of [0, 800] m and on the other side of a major road to urban parks attract more visitors than hexagons located in the walking distance band of [800, 1000] m and on the other side of a major road, on average. Moreover, hexagons with EA to urban parks experience an overall increase in visits of 1.0%. That is, on average, the difference in daily visits to hexagons with EA to urban parks compared to hexagons in the 0–800 m walking distance band and on the other side of a major road is 1.0% higher than the difference in daily visits to hexagons in the 1000–1800 m walking distance band of an urban park and on the park side of a major road compared to hexagons in the 1000–1800 m walking distance band and on the other side of a major road. Given the mean value of 1719.71 daily trips per hexagon with EA to urban parks (hexagons in area A) and the sample weight of these cellphone data to the total population in Shenzhen, the increased number of daily trips to a hexagon on the park side of a main road is approximately $(1719.71 / 0.4) \times 0.01 = 43$. The increase in daily visits due to EA to urban parks is economically significant, compared with the mean population density in Shenzhen.²⁰

Columns (2) – (3) further divide the sample according to park type. As expected, hexagons with EA to city parks have more visits, as measured by anonymous cellphone data. On average, the difference in daily visits to hexagons with EA to city parks compared to hexagons in the 0–800 m walking distance band and on the other side of a major road is 1.2% higher than the difference in daily visits to hexagons in the 1000–1800 m walking distance band of the same park and on the park side of a major road compared to hexagons in the 1000–1800 m walk band and on the other side of a major road. Moreover, the coefficient of the interaction term of $(D \times \text{Access})$ in Column (2), which measures changes in the daily visits to hexagons with EA to neighborhood parks, is only marginally significant at the 10% level. That is, the number of daily visits to hexagons with EA to neighborhood parks decreases by 9.34 trips, which is marginally significant at the 10% level, while the number of daily visits to hexagons with EA to city parks increases by approximately 56 trips, which is statistically significant at the 1% level.²¹

Overall, these findings provide support for our argument that the areas on the city park side of a major road experience a greater influx of visitors, resulting in disamenities and, therefore, negative effects on houses with EA to the city parks.

6. Conclusion

In this paper, we separate the costs of proximity to urban parks from the combined effects by using a boundary discontinuity approach. We

¹⁸ Post_10 equals 1 if a house is sold after 10 or more years after the park is built.

¹⁹ All parks in the sample were open before 2019. Due to the short time span of the cellphone data used in this study, we drop the time-varying dummy of *Open*, and the triple difference estimator accordingly.

²⁰ The mean value of daily trips per hexagon with EA to urban parks is 1719.71 per hexagon, and the population density per hexagon in Shenzhen is 437.5 people per hexagon, approximately 7000 people per square kilometer.

²¹ The mean value of daily trips per hexagon with EA to neighborhood parks and city parks are 1246.20 and 1815.13, respectively. Therefore, the changes in daily trips per hexagon with EA to neighborhood parks is $(1246.20 \div 40\%) \times (-0.003) = -9.34$ trips, and the changes in daily trips per hexagon with EA to city parks is $(1866.40 \div 40\%) \times 0.012 = 55.99$ trips.

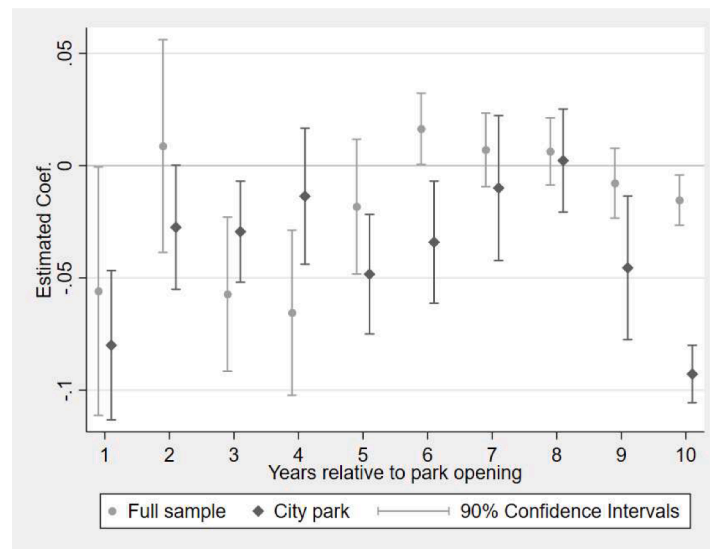


Fig. 4. The dynamic effects of EA on housing prices.

Note: This figure plots the dynamic effects of EA on housing prices, estimated by the interaction $D \times Access \times Post_t$, which equals 1 if a house is sold after t years of park opening. The estimations use the full samples and houses close to the city park separately, located within the 1800 distance band. All the regressions control hedonic relative variables, park fixed effects, town $\times$ year fixed effects, and quarter fixed effects. The dots represent the estimated coefficients, while the bar lines represent the 90% confidence intervals. The estimated results are also presented in Table B2 in Appendix B.

Table 7

Mechanism: EA of a hexagon to a park increases daily visits.

	Full (1)	Neighborhood park (2)	City park (3)
D	0.001*** (0.001)	0.008*** (0.001)	0.002*** (0.001)
Access	-0.011*** (0.001)	-0.011*** (0.001)	-0.013*** (0.001)
D $\times$ Access	0.010*** (0.001)	-0.003* (0.002)	0.012*** (0.001)
Home	0.001 (0.001)	0.011*** (0.001)	-0.002*** (0.0001)
Work	0.016*** (0.001)	0.001*** (0.001)	0.019*** (0.001)
Dis. to major road	0.002*** (0.001)	0.001*** (0.001)	0.002*** (0.001)
Observations	22,524,063	1,663,723	20,860,340
R-squared	0.013	0.010	0.013
Park FE	Yes	Yes	Yes

This table reports the mechanism analysis of the negative effects of EA caused by the high volume of daily visits. The dependent variable is the logarithmic value of the daily mean number of visitors in hexagon pairs in October 2019, which is measured by anonymous cellphone data. Only the destination hexagons within the walking distance band of [0, 800] m and the control band of [1000, 1800] m to urban parks are included. Column (1) uses the full sample, and Columns (2) – (3) separate the full samples based on the type of nearest park, including neighborhood parks and city parks. The dummy D takes a value of one if destination hexagon h located within 800 m walking distance of the nearest park, and zero if destination hexagon h is within 1000–1800 m walking distance of the nearest park. $Access$ is a time-constant dummy that takes a value of one if the destination hexagon h located on the park side of the major road at the end of the study period, and zero otherwise. $Home$ is the logarithmic value of the number of individuals living in a given destination grid, while $Work$ is the logarithmic value of the number of individuals working in a given destination. Park fixed effects are included. Robust standard errors are in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

find significant negative effects of EA to urban parks on housing prices. On average, houses with EA to urban parks experience a price discount of 2.0%, equivalent to approximately \$7,986 for houses of average value. The results are robust when we exclude samples that may cause confounders. We also find suggestive evidence that residents do not systematically sort across the border. However, we find that houses are more sensitive to the disamenities caused by EA to city parks. We also show that the negative effect of EA to urban parks is consistent over time.

We argue that one of the disamenities of EA to urban parks is caused by the high volume of visits to the areas on the park side of a major road. Using novel cellphone data, we find that areas on the park side of a major road are more visited, resulting in disamenities such as noise,

traffic congestion, and privacy loss. From a policy perspective, our evidence provides new insights into the negative externalities of proximity to urban parks. This knowledge should be considered in the design of urban land use planning.

CRediT authorship contribution statement

Lu Liu: Data curation, Writing – original draft, Visualization. **Lina Meng:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Ruige Zhang:** Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare no conflict of interest related to the submitted manuscript.

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Appendix A. Additional data descriptions



Fig. A1. The conceptual graph of a major road.
Note: It is a conceptual graph of a major road drawn based on *The Guide of Road Design in Shenzhen*, approved by the Housing and Construction Bureau of Shenzhen Municipality. ① The street buildings; ② the sidewalk; ③ the street tree separating the auxiliary lane and the trunk road ④; ⑤ the strips that act as a physical barrier to pedestrians who want to cross the major road.



Fig. A2. Images view of parks: (A) neighborhood park; (B) city park.
Note: The screenshot of park (A) is Shenzhen Shenwan Street Park, which was acquired from ArchDaily. The screenshot of park (B) is Shenzhen Central Park which was acquired from the Shenzhen News Website.

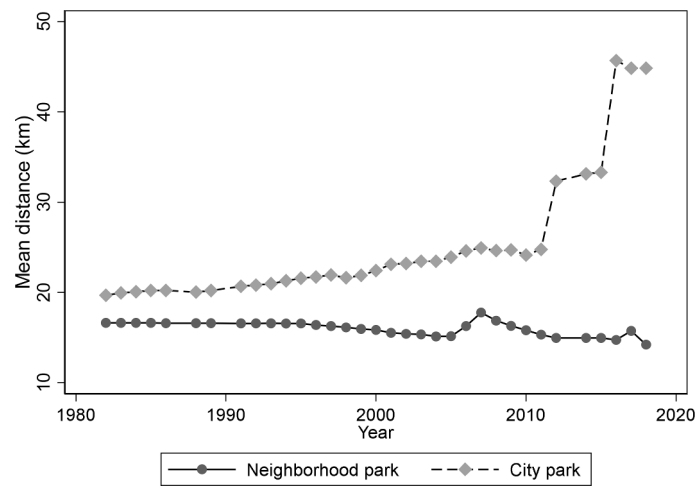


Fig. A3. The mean distance of urban parks to the city center in Shenzhen over time.

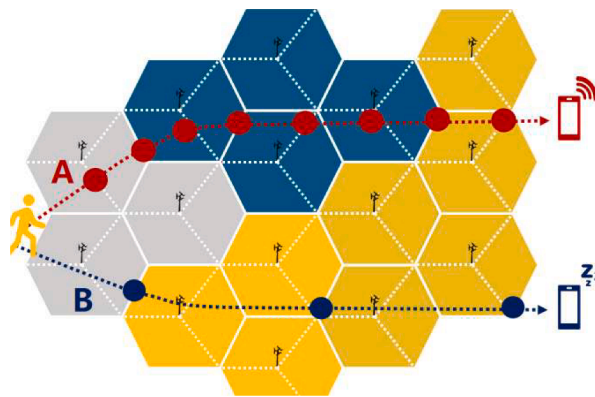


Fig. A4. Measurement of people's mobility by anonymous cellphone data.

This figure shows how cellphone data are used to identify individuals' mobility through signal transmission among base stations. The dataset is calibrated by a series of algorithms on the raw data. Simply, each hexagon has a base station. A cellphone connects to its nearest base station for data transformation. People A is tagged as moving across hexagon h if his or her cellphone telecommunicates with the base station in hexagon h within a certain time slot. One is tagged as staying in hexagon h if the telecommunication time exceeds the threshold. See <http://www.smartsteps.com/index.php> for more information on the data introduction.

Table A1

Descriptive statistics of housing transactions in Shenzhen from 2005 to 2018.

	N	Mean	S.D.	P1	P99
Price (thousand \$)	219,761	396.82	367.76	45.93	2159.75
Dist. to the nearest parks (m)	219,763	1019.00	587.90	45.09	3004.00
Access	217,281	0.63	0.48	0	1
Open	187,979	0.94	0.23	0	1
Area of the nearest park (ha)	208,427	11.29	1.61	9.22	15.07
No. of parks	219,220	0.25	0.52	0	2
Age of house	208,427	10.61	5.76	1	27
Living areas (m ²)	219,763	79.17	35.02	27.78	225.40
No. bedroom	195,332	2.24	0.96	1	5
Area of residential complex (ha)	204,624	36343.02	43780.22	2108.36	267851.50
Primary school level	219,763	1.14	1.42	0	4
Dist. to CBD (km)	219,763	11.43	7.48	0.04	33.19
Dist. to mall (km)	219,763	1.52	1.65	0.07	9.52
Dist. to metro station (km)	219,763	1.73	2.80	0.06	13.49
Dist. to coastlines (km)	219,763	6.91	4.94	0.28	17.02
Dist. to hospital (km)	219,763	1.78	1.41	0.12	6.63
Dist. to major road (m)	219,763	194.02	156.69	33.91	373.74

This table reports summary statistics of housing transactions in Shenzhen from 2005 to 2018 using the full sample.

Table A2

Descriptive statistics by construction year of parks.

Park's Built Year	Neighborhood Park Count	Area (ha)		City Park Count	Area (ha)	
		Mean	S.D.		Mean	S.D.
Before 2005	29	9.97	0.48	38	12.49	1.44
2005 - 2009	35	10.11	0.73	11	12.32	1.41
2010 - 2014	5	10.19	0.83	17	12.69	1.60
2015 - 2018	12	10.00	0.75	4	11.46	1.13

Table A3

Descriptive statistics by different park types.

	Neighborhood Park		City park	
	Mean	S.D.	Mean	S.D.
Price (thousand \$)	389.35	337.47	388.58	351.86
D	0.54	0.50	0.53	0.50
Access	0.70	0.46	0.59	0.49
Open	0.96	0.19	0.93	0.25
Dist. to the nearest parks (m)	843.17	474.65	892.55	460.45
Park areas (ha)	10.10	0.58	12.54	1.42
Age of house	11.00	5.59	10.68	5.44
Living areas (m ²)	76.52	31.31	78.25	32.91
No. bedroom	2.23	0.93	2.28	0.92
Residential complex (ha)	2.73	3.29	2.79	3.41
Primary school level	1.32	1.51	1.23	1.48
Dist. to CBD (km)	10.80	6.81	9.97	6.75
Dist. To mall (km)	1.13	1.10	1.65	1.72
Dist. to metro station (km)	1.47	2.47	1.59	2.51
Dist. to coastlines (km)	5.42	4.11	6.01	4.56
Dist. to hospital (km)	1.35	0.89	1.55	1.19
Dist. to major road (m)	174.61	119.61	193.45	164.57
Observation	38540		49403	

Appendix B. Additional Results

B.1. Parallel trends test by event study

To verify the parallel trends of housing prices with EA before and after park opening, we implement an event study as follows:

$$\log P_{ij,t} = \beta_0 + \sum_{g=-5, g \neq -1}^{g=5} \lambda_g [Access_{ij} \times I(t - Open_k)] + \sum_{g=-5, g \neq -1}^{g=5} \tau_g [D_{ij} \times Access_{ij} \times I(t - Open_k)] + \beta_1 D_{ij} + \beta_4 Access_{ij} + \beta_5 (D_{ij} \times Access_{ij}) + \rho X_{ij} + \eta_t \times v_c + k_{j,park} + q_{qt} + \varepsilon_{ijt} \quad (B.1)$$

where $Open_k$ is the opening year of nearest park k , and $I(\cdot)$ is an indicator that equals one when $t - Open_k = g$ and zero otherwise. $Access_{ij}$ is a time-constant dummy, which takes a value of one if house i is located on the park side of the major road at the end of the study period. All other variables' definitions are the same as in Eq. (2). Then, the coefficients $\tau_{-5}, \dots, \tau_{-2}$ measure the relative pretrends for $D_{ij} = 1$ versus $D_{ij} = 0$, and $\lambda_{-5}, \dots, \lambda_{-2}$ are the pretrends for $D_{ij} = 0$.

This analysis includes all houses located within 1800 m of the nearest parks. To eliminate the spillover effects, we exclude houses in a distance band of (800, 1000) m. We further exclude house transactions before the major road construction. Taking one year before park opening ($g = -1$) as a reference, the results in Fig. B1 show that, overall, the coefficients of τ_{-5} to τ_{-2} and λ_{-5} to λ_{-2} are statistically insignificant, providing suggestive evidence that the parallel trends of housing prices between houses with EA and those without EA hold before the park opens.

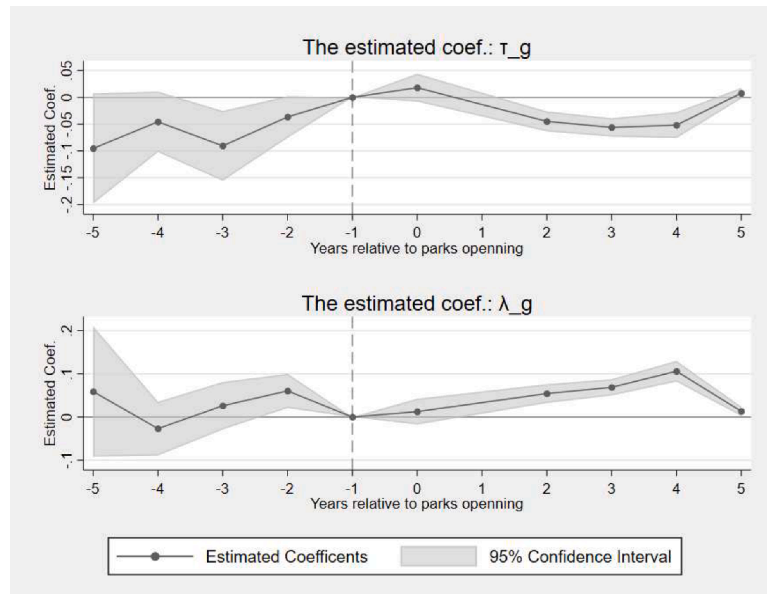


Fig. B1. Event study estimations.

Note: This figure illustrates the yearly changes in housing prices with EA before and after the nearest park opening. The dots indicate point estimations, and the gray areas indicate the 95% confidence intervals.

B.2. The dynamic effects of EA on housing prices

Table B2

This table reports summary statistics of housing transactions in Shenzhen from 2005 to 2018 using the full sample.

	(1) Full sample	(2) City park
D × Access × Post_1	-0.056* (0.034)	-0.062*** (0.019)
D × Access × Post_2	0.009 (0.029)	-0.025 (0.017)
D × Access × Post_3	-0.057*** (0.021)	-0.031** (0.014)
D × Access × Post_4	-0.066*** (0.022)	-0.005 (0.018)
D × Access × Post_5	-0.018 (0.018)	-0.027* (0.016)
D × Access × Post_6	0.016* (0.010)	-0.016 (0.017)
D × Access × Post_7	0.007 (0.010)	0.001 (0.019)
D × Access × Post_8	0.006 (0.009)	0.006 (0.014)
D × Access × Post_9	-0.008 (0.009)	-0.022 (0.020)
D × Access × Post_10	-0.015** (0.007)	-0.056*** (0.009)
D	0.010* (0.006)	0.033*** (0.008)
Access	-0.010 (0.011)	0.019 (0.016)
Controls	Yes	Yes
Park FE	Yes	Yes
Town × year FE	Yes	Yes
Quarter FE	Yes	Yes
N	87883	49376
adj. R ²	0.904	0.909

The dynamic effects of EA on housing prices. This table reports the dynamic effects of EA on housing prices, which is the same as shown in Fig. 4. The sample comprises all houses located within an 1800 m walking distance of the nearest park. The dependent variable is the logarithmic form of the transaction price of houses. The dummy D takes a value of one if a house is located within an 800 m walking distance of the nearest park, and zero if the house is within a 1000–1800 m walking distance of the nearest park. $Access$ is a dummy which takes a value of one if a house is located on the park side of a major road, within a walking distance of 0–1800 m of the nearest park. $Post\ t$ is a dummy variable that equals 1 if a house has been sold since t years of park construction. All the regressions control hedonic relative variables, park fixed effects, town $\times$ year fixed effects, and quarter fixed effects. Robust standard errors in parentheses are clustered by town. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.3. The effects of proximity to the nearest park on visits

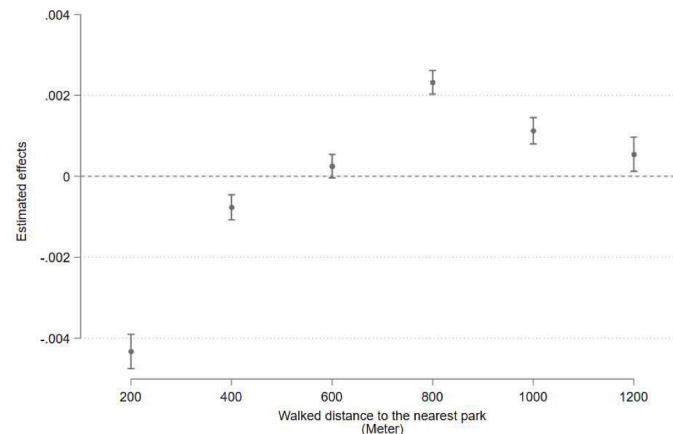


Fig. B2. The effects of proximity to the nearest park on visits.

Note: This figure depicts the effects of proximity to the nearest parks on visits. The sample used in the analysis is all sales after park opening and within a walking distance of [0, 2400] m of the nearest park. Each dot is based on a separate regression of housing prices on a walked distance band of d meters, using houses within a walking distance band of [1200, 2400] as controls. The vertical lines represent 90% confidence intervals. We control the logarithmic value of people working in the hexagon, the logarithmic value of people living in the hexagon, and the distance to the major road. Park fixed effects are included in each regression.

References

- Albouy, David, Christensen, Peter, Sarmiento-Barbieri, Ignacio, 2020. Unlocking Amenities: Estimating Public Good Complementarity. *Journal of Public Economics* 182, 104110.
- Angelovski, Isabelle, Cole, Helen, Connolly, James, Triguero-Mas, Margarita, 2018. Do Green Neighbourhoods Promote Urban Health Justice? *The Lancet Public Health* 3 (6), e270.
- Bakkensen, Laura A., Ma, Lala, 2020. Sorting over Flood Risk and Implications for Policy Reform. *Journal of Environmental Economics and Management* 104, 102362.
- Bayer, Patrick, Ferreira, Fernando, McMillan, Robert, 2007. A Unified Framework for Measuring Preferences for Schools and Neighborhoods. *Journal of Political Economy* 115 (4), 588–638.
- Bernstein, Asaf, Gustafson, Matthew T., Lewis, Ryan, 2019. Disaster on the Horizon: The Price Effect of Sea Level Rise. *Journal of Financial Economics* 134 (2), 253–272.
- Bertram, Christine, Rehdanz, Katrin, 2015. The Role of Urban Green Space for Human Well-Being. *Ecological Economics* 120, 139–152.
- Bibler, Andrew, Billings, Stephen B., 2020. Win or Lose: Residential Sorting after a School Choice Lottery. *The Review of Economics and Statistics* 102 (3), 457–472.
- Black, Sandra E., 1999. Do Better Schools Matter? Parental Valuation of Elementary Education. *The Quarterly Journal of Economics* 114 (2), 577–599.
- Bosker, Maarten, Garretsen, Harry, Marlet, Gerard, Woerkens, Clemens van, 2018. Nether Lands: Evidence on the Price and Perception of Rare Natural Disasters. *Journal of the European Economic Association* 17 (2), 413–453.
- Chan, Jimmy, Fang, Xian, Wang, Zhi, Zai, Xianhua, Zhang, Qinghua, 2020. Valuing Primary Schools in Urban China. *Journal of Urban Economics* 115, 103183.
- Christensen, Peter, Sarmiento-Barbieri, Ignacio, Timmins, Christopher, 2022. Housing Discrimination and the Toxics Exposure Gap in the United States: Evidence from the Rental Market. *The Review of Economics and Statistics* 104 (5), 807–818.
- Chu, Junhong, Duan, Yige, Yang, Xianling, Wang, Li, 2020. The Last Mile Matters: Impact of Dockless Bike Sharing on Subway Housing Price Premium. *Management Science* 67 (1), 297–316.
- Cohen, Jeffrey P., Danko, Joseph J., Yang, Ke, 2019. Proximity to a Water Supply Reservoir and Dams: Is There Spatial Heterogeneity in the Effects on Housing Prices? *Journal of Housing Economics* 43, 14–22.
- Currie, Janet, Davis, Lucas, Greenstone, Michael, Walker, Reed, 2015. Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings. *American Economic Review* 105 (2), 678–709.
- Daams, Michiel N., Sijtsma, Frans J., Vlist, Arno J. van der, 2016. The Effect of Natural Space on Nearby Property Prices: Accounting for Perceived Attractiveness. *Land Economics* 92 (3), 389–410.
- Fan, Qin, Andrew Hansz, J., Yang, Xiaoming, 2016. The Pricing Effects of Open Space Amenities. *The Journal of Real Estate Finance and Economics* 52 (3), 244–271.
- Franco, Sofia F., Macdonald, Jacob L., 2018. Measurement and Valuation of Urban Greenness: Remote Sensing and Hedonic Applications to Lisbon, Portugal. *Regional Science and Urban Economics* 72, 156–180.
- Gibbons, Stephen, 2015. Gone with the Wind: Valuing the Visual Impacts of Wind Turbines through House Prices. *Journal of Environmental Economics and Management* 72, 177–196.
- Gibbons, Stephen, Machin, Stephen, Silva, Olmo, 2013. Valuing School Quality Using Boundary Discontinuities. *Journal of Urban Economics* 75, 15–28.
- Gibbons, Stephen, McNally, Sandra, Viarengo, Martina, 2017. Does Additional Spending Help Urban Schools? An Evaluation Using Boundary Discontinuities. *Journal of the European Economic Association* 16 (5), 1618–1668.
- Greenstone, Michael, Gallagher, Justin, 2008. Does Hazardous Waste Matter? Evidence from the Housing Market and the Superfund Program. *The Quarterly Journal of Economics* 123 (3), 951–1003.
- Jones, Benjamin A., Goodkind, Andrew L., 2019. Urban Afforestation and Infant Health: Evidence from Milliontreesnyc. *Journal of Environmental Economics and Management* 95, 26–44.
- Kuroda, Yuta, Sugawara, Takeru, 2023. The Value of Scattered Greenery in Urban Areas: A Hedonic Analysis in Japan. *Environmental and Resource Economics* 85, 523–586.
- Kuwayama, Yusuke, Olmstead, Sheila, Zheng, Jiameng, 2018. The Value of Water Quality: Separating Amenity and Recreational Benefits. <https://ostromworkshop.indiana.edu/pdf/seriespapers/2018spr-colloq/olmstead-paper.pdf>.
- Liu, Zhaoyang, Hanley, Nick, Campbell, Danny, 2020. Linking Urban Air Pollution with Residents' Willingness to Pay for Greenspace: A Choice Experiment Study in Beijing. *Journal of Environmental Economics and Management* 104, 102383.
- Michaels, Guy, Nigmatulina, Dzhamilya, Rauch, Ferdinand, Regan, Tanner, Baruah, Neeraj, Dahlstrand, Amanda, 2021. Planning Ahead for Better

- Neighborhoods: Long-Run Evidence from Tanzania. *Journal of Political Economy* 129 (7), 2112–2156.
- Muehlenbachs, Lucija, Spiller, Elisheba, Timmins, Christopher, 2015. The Housing Market Impacts of Shale Gas Development. *American Economic Review* 105 (12), 3633–3659.
- Panduro, Toke Emil, Jensen, Cathrine Ulla, Lundhede, Thomas Hedemark, Graevenitz, Kathrine von, Thorsen, Bo Jellesmark, 2018. Eliciting Preferences for Urban Parks. *Regional Science and Urban Economics* 73, 127–142.
- Rosen, Sherwin., 1974. Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition. *Journal of Political Economy* 82 (1), 34–55.
- Rouwendal, Jan, Levkovich, Or, Marwijk, Ramona van, 2017. Estimating the Value of Proximity to Water, When Ceteris Really Is Paribus. *Real Estate Economics* 45 (4), 829–860.
- Tyrväinen, Liisa, Miettinen, Antti, 2000. Property Prices and Urban Forest Amenities. *Journal of Environmental Economics and Management* 39 (2), 205–223.
- von Graevenitz, Kathrine., 2018. The Amenity Cost of Road Noise. *Journal of Environmental Economics and Management* 90, 1–22.
- Walls, Margaret, Kousky, Carolyn, Chu, Ziyang, 2015. Is What You See What You Get? The Value of Natural Landscape Views. *Land Economics* 91 (1), 1–19.